- *Backend as a Service* (BaaS)—In this model, applications rely heavily on third-party services offered by cloud providers, such as databases, authentication services, and message queues. The focus is on reducing development and operational costs related to backend services. Developers can implement frontend applications (such as single-page applications or mobile applications) while offloading most or all of the backend functionality to BaaS vendors. For example, they could use Okta to authenticate users, Google Firebase for persisting data, and Amazon API Gateway to publish and manage REST APIs.
- *Function as a Service* (FaaS)—In this model, applications are stateless, triggered by events, and fully managed by the platform. The focus is on reducing deployment and operations costs related to orchestrating and scaling applications. Developers can implement the business logic for their applications, and the platform takes care of the rest. Serverless applications don't have to be implemented with functions to be categorized as such. There are two main FaaS offerings. One option is to go with vendor-specific FaaS platforms, such as AWS Lambda, Azure Functions, or Google Cloud Functions. Another option is to choose a serverless platform based on open source projects, which can run either in a public cloud or on premises, addressing concerns like vendor lock-in and lack of control. Examples of such projects are Knative and Apache OpenWhisk. Knative provides a serverless runtime environment on top of Kubernetes, as you'll see in chapter 16. It's used as the foundation for enterprise serverless platforms like VMware Tanzu Application Platform, RedHat OpenShift Serverless, and Google Cloud Run.